

Behavioral Cues for Trust in AI-Driven Virtual Agents

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ABSTRACT

Artificial Intelligence (AI) is increasingly integrated into immersive environments, accelerating the use of virtual agents. Yet it remains unclear which interaction characteristics shape users' perceptions of an AI-driven agent's trustworthiness. We conducted a controlled user study with 30 participants to evaluate how different animations and response-time delays influence perceived trust in virtual agents in a Virtual Reality (VR) environment. Our results show that agent animation, response-time delay, and response complexity jointly influence perceived trust in AI-driven agents, with attentive animation increasing trust and delayed responses reducing it.

Index Terms: HMD, anthropomorphism, trust

1 INTRODUCTION

In rapidly changing societies, Artificial Intelligence (AI) has become increasingly prevalent across interactive systems. In particular, human-agent interaction in Virtual and Augmented Reality (VR/AR) increasingly leverages AI to support adaptive behavior and context-aware responses [3]. These AI-driven capabilities enable more responsive, and scalable interactions between users and computational agents within immersive environments.

AI-driven virtual agents, however, do not always exhibit human-like responses. They may respond to queries with unnaturally short delays or take excessive time to answer simple or complex questions, producing response timings that can undermine users' perceived trust and the naturalness of the interaction. These unnatural response patterns may negatively affect user perceptions, such as trust. Prior work shows that moderate response delays can enhance perceived naturalness, whereas longer delays degrade user experience [6]. To mitigate perceived latency, researchers have explored behavioral strategies, including semantically aligned agent animations, which have been shown to reduce perceived response time, particularly under high-delay conditions [4]. Users also tolerate longer delays when agents provide more complex information [3], suggesting that perceived response time depends on both timing and response characteristics. Despite these findings, it remains unclear whether previously reported effects of response timing and agent behavior generalize to substantially longer response delays, such as those introduced by AI-driven systems, and across different answer structures. This work addresses this gap by examining response timing and animation strategies under long response delays in virtual agents, informing the design of AI-driven agents that support natural and trustworthy interactions in immersive environments.

2 USER STUDY DESIGN

This study investigates how AI-driven virtual agent behavior and response timing influence users' perceived trust during question-answer interactions in VR, addressing the research question:

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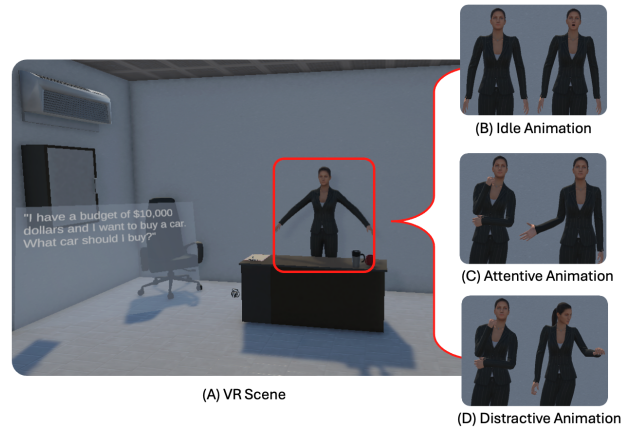


Figure 1: (A) shows the VR scene. The virtual agent and three animations, including (B) Idle, (C) Attentive, and (D) Distracted. For each animation, the agent first displays the left behavior during the response delay and then displays the right behavior while speaking.

“How do response-time delay, agent animation behavior, and response length affect users' trust in an AI-driven virtual agent?”

2.1 Study Task and Conditions

We designed a controlled VR-based question-answer task in which participants interacted with a virtual agent under systematically varied conditions. The experiment manipulated three factors: response-time delay, agent animation behavior, and response complexity. Response-time delay was set to 1 second, 10 seconds, or 20 seconds (*RT1*, *RT10*, and *RT20*). Agent animation included three behaviors: *Attentive*, *Distracted*, and *Idle* (Figure 1 B-D). Response complexity was manipulated at two levels, corresponding to *Short-Answer* (fact-based, concise) or *Long-Answer* (explanatory, comparative) responses. A representative short-answer question was “Exactly how far is it from the Eiffel Tower to the Colosseum in Rome?” A representative long-answer question was “I am deciding whether to buy a MacBook or a Dell laptop. Can you explain the advantages and disadvantages of each?” All questions appeared in a prompt window, allowing participants to read and verbally ask them. The design produced 18 scenarios. Across conditions, participants interacted with the same virtual agent in an identical environment to control visual and contextual factors. The agent was consistently presented as a Caucasian American woman.

In each trial, participants verbally asked a predefined question, and the agent responded after a delay with an assigned animation. The *Attentive* animation showed eye contact and subtle head movements, *Distracted* showed environmental scanning and unrelated motions, and *Idle* showed no visible movement during the delay.

2.2 AI-driven Virtual Agent

The virtual agent's verbal responses were generated using Groq's production-level large language model, LLaMA-3.3-70B-

Versatile [2]. The AI-generated responses supported a realistic and ecologically valid simulation of contemporary user–AI interactions.

We selected a Caucasian American female agent (Fig. 1) with an American accent, reflecting common design conventions in widely deployed virtual assistants such as Siri and Alexa [5]. Because the participant pool primarily consisted of U.S. university students, this representation was expected to be familiar and to minimize potential confounding effects related to voice or appearance. The agent model and animations were sourced from the Microsoft Rocketbox repository [1]. The agent wore business-casual attire, and the virtual environment resembled a professional office setting. Three animation conditions were implemented to systematically vary perceived engagement and naturalness during response delays.

2.3 Participants, Apparatus, and Study Procedures

We had 30 participants (17 males and 13 females, average age: 19.4) in the study (IRB #15385). Twenty-seven of the participants indicated that they are working on their undergraduate degree.

To begin, the participant put on the Meta Quest 3 and went through a short training on how to interact with the virtual agent. The virtual scene for the study was developed in Unity version 6000.1.4f1. They then entered the prepared scene, asked the pre-prepared question, listened to the virtual agent's response, and completed the in-study questionnaires. This process was repeated in virtual reality until all 18 scenarios were completed. To conclude the study, the participant completed a post-questionnaire to provide feedback on the perceived trust of the virtual agent they interacted with. The entire study was completed in under 60 minutes.

2.4 Measurements and Analysis

We employed two subscales, *Sincere* and *Authentic*, from the Multi-Dimensional Measure of Trust (MDMT)[7] to assess participants' perceived trust-related perceptions of the virtual agent. *Sincere* captures whether the agent's responses are perceived as genuine and well-intentioned, while *Authentic* assesses whether the agent is perceived as natural, human-like, and true to itself. These two dimensions directly reflect users' moment-to-moment judgments of an agent's behavioral credibility and interaction naturalness, which are central to trust formation in conversational and agent interactions. All measures used a 7-point Likert scale. We analyzed data with repeated-measures ANOVA at a 5% significance level. Greenhouse–Geisser corrections addressed sphericity violations, and significant effects were followed by Bonferroni-corrected pairwise comparisons.

3 RESULTS

Sincere: A significant interaction effect was observed between animation and response complexity ($F(1.528, 44.312)=4.079$, $p=.033$, $\eta_p^2=.123$). Follow-up simple effects analyses revealed a significant effect of animation for *Short-Answer* responses ($F(2,28)=5.156$, $p=.012$, $\eta_p^2=.269$). Both *Attentive* (4.033 [3.556, 4.511], $p=.01$) and *Distracted* (4.022 [3.48, 4.564], $p=.015$) yielded higher scores than *Idle* (3.433 [2.775, 4.091]), with no significant difference between *Attentive* and *Distracted* ($p=1.00$).

A significant simple effect of animation was also found for *Long-Answer* responses ($F(2, 58)=7.651$, $p=.002$, $\eta_p^2=.353$). In this condition, *Attentive* (4.8 [4.307, 5.293]) resulted in higher scores than both *Distracted* (4.278 [3.71, 4.846], $p=.006$) and *Idle* (4.2 [3.643, 4.757], $p=.003$).

There is a main effect on animation ($F(1.368, 39.67) = 11.518$, $p<.001$, $\eta_p^2=.284$). *Attentive* (4.417 [3.947, 4.886]) showed a higher score than *Distracted* (4.15 [3.632, 4.668], $p=.014$) and *Idle* (3.817 [3.237, 4.396], $p=.002$). We also observed *Distracted* was higher than *Idle* ($p=.025$). A main effect is also found on response delay ($F(2, 58)=10.786$, $p<.001$, $\eta_p^2=.271$). *RT1* (4.439 [3.869, 5.009]) did not differ significantly from *RT10* (4.161

[3.663, 4.659] ($p=.29$), differed significantly from *RT20* (3.783 [3.263, 4.304] ($p<.001$). *RT10* showed higher scores than *RT20* ($p=.02$). Furthermore, there is a main effect on response complexity ($F(1, 29)=24.447$, $p<.001$, $\eta_p^2=.457$). *Long-Answer* (4.426 [3.918, 4.934], $p<.001$) was higher than *Short-Answer* (3.830 [3.302, 4.357]).

Authentic: We found a significant interaction effect between the response delay and complexity ($F(2, 58)=4.619$, $p=.014$, $\eta_p^2=.137$). A significant simple effect was observed for *Short-Answer* ($F(2, 58)=9.708$, $p<.001$, $\eta_p^2=.251$) and *Long-Answer* ($F(2, 58)=4.720$, $p=.013$, $\eta_p^2=.14$). For both types, we observed similar results, showing *RT1* (short-answer: 4.2 [3.553, 4.847], long-answer: 4.289 [3.696, 4.882]) had significantly higher authenticity than *RT20* (short-answer: 3.344 [2.789, 3.9], $p<.001$, long-answer: 3.767 [3.247, 4.287], $p=.009$).

There is a main effect on animation ($F(2,58)=14.093$, $p<.001$, $\eta_p^2=.327$). *Attentive* (4.289 [3.804, 4.774]) showed a higher score than *Distracted* (3.879 [3.36, 4.396], $p=.004$) and *Idle* (3.611 [3.062, 4.161], $p<.001$). A main effect is also found on response delay ($F(2,58)=7.878$, $p<.001$, $\eta_p^2=.214$). *RT1* (4.244 [3.642, 4.847]) showed a higher score than *RT20* (3.556 [3.037, 4.074], $p<.001$). Furthermore, a main effect of response complexity was observed ($F(1,29)=12.459$, $p=.001$, $\eta_p^2=.301$), with *Long-Answer* (4.115 [3.633, 4.596]) receiving higher authenticity than *Short-Answer* (3.737 [3.205, 4.269], $p=.001$).

4 DISCUSSION AND CONCLUSION

Our results show that agent behavior, response delay, and answer type significantly influence users' trust in AI-driven virtual agents in VR. *Attentive* animation increased perceived sincerity, while longer delays reduced trust, indicating sensitivity to latency. Animation benefits were strongest for long answers, suggesting users expect visible cognitive engagement during complex responses. Longer responses were perceived as more trustworthy, partially offsetting delay effects. Overall, attentive animations, shorter delays, and richer responses support more trustworthy agent interactions.

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